Algorithm for file updates in Python

Project description

I am tasked with regular updating of the file that identifies the employees who can access restricted content on a subnetwork. There is an allow list for employee IP addresses with access. There is also a remove list for IP addresses that identifies employees that I must remove from the allow list. I must create a Python algorithm script to check the list and remove IP addresses from the file.

Open the file that contains the allow list

```
# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:
```

Read the file contents

```
# Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

  ip_addresses = file.read()
```

Convert the string into a list

```
# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()
```

Iterate through the remove list

```
# Build iterative statement
 # Name loop variable `element`
 # Loop through `ip_addresses`

 for element in ip_addresses:

   # Build conditional statement
   # If current element is in `remove_list`,

   if element in remove_list:
```

Remove IP addresses that are on the remove list

```
# then current element should be removed from `ip_addresses`

     ip_addresses.remove(element)
```

Update the file with the revised list of IP addresses

```
# Use `.join()` to turn `ip_addresses` back into a string for file
writing

 ip_addresses = "\n".join(ip_addresses)

 # Build `with` statement to overwrite the target file

 with open(import_file, "w") as file:
```

```
    # Use `.write()` to append updated string data back into the
file

    file.write(ip_addresses)
```

Summary

The algorithm reads the existing allow list file and uses `.read()` to store its contents in a variable. It then uses `.split()` to convert the IP addresses from a string into a list. Next, it iterates through the list and checks whether each IP address appears in the remove list. If an IP address is found in the remove list, the algorithm removes it from the allow list. Finally, it uses `.join()` to convert the updated list back into a string and `.write()` to overwrite the original file with the revised list of authorized IP addresses.